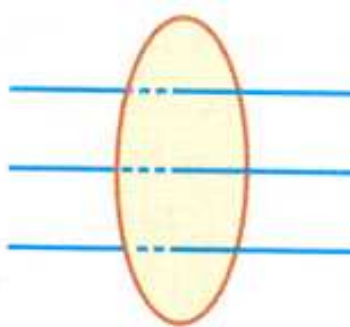


assignment 9

sb-prob3161a.problem

Due date: Sun Mar 22 11:59:59 pm 2026 (EDT)

A circular loop of wire of radius 8.18cm is in a uniform magnetic field, with the plane of the loop perpendicular to the direction of the field. The magnetic field varies with time according to $B(t) = 0.060 + 2.0t$, where t is in seconds, and B is in T. Calculate the magnetic flux through the loop at $t = 0$ s.



Tries 0/10

Calculate the magnitude of the emf induced in the loop.

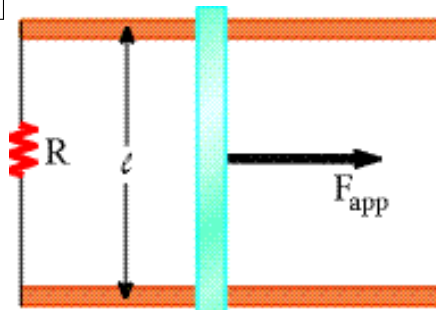
Tries 0/10

Calculate the magnitude of the induced current in the loop if the resistance in the wire is 0.114ohm.

Tries 0/10

sb-prob3122a.problem

A conducting rod of length l moves on two horizontal, frictionless rails, as shown in the figure below. If a constant force of 1.66N is needed to move the bar at constant speed 2.16m/s through a magnetic field $\mathbf{B}$ that is directed into the page, what is the current through the 8.04 Ω resistor R ?



Tries 0/10

What is the rate at which energy is delivered to the resistor?

Tries 0/10

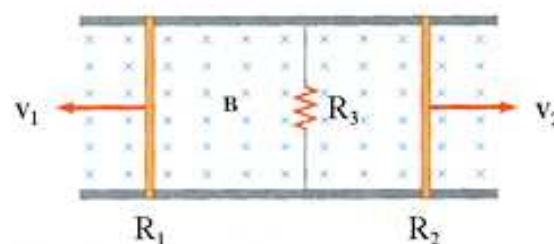
What is the mechanical power delivered by the force $\mathbf{F}_{app}$?

Tries 0/10

sb-prob3131.problem

Two parallel rails with negligible resistance are 1.00E-1m apart and are connected by a 3.70 Ω resistor. The circuit also contains two metal rods having resistances of $R_1=8.00\Omega$ and $R_2=14.90\Omega$ sliding along the rails (see figure). The rods are pulled away from the resistor at constant speeds $v_1=4.00\text{m/s}$ and $v_2=2.00\text{m/s}$, respectively. A uniform magnetic field of magnitude 0.0100T is applied perpendicular to the plane of the rails. Determine the current in the 3.70ohm resistor.

Tries 0/10



sb-prob3158.problem

Magnetic field values are often determined by using a device known as a search coil. This technique depends on the measurement of the total charge passing through a coil in a time interval during which the magnetic flux linking the windings changes either because of the motion of the coil or because of a change in the value of B . As a specific example, calculate B when a 57-turn coil of resistance 167 Ω and cross-sectional area 35.4m² produces the following results: A total charge of 5.24E-4C passes through the coil when it is rotated in a uniform field from a position where the plane of the coil is perpendicular to the field to a position where the coil's plane is parallel to the field.

Tries 0/10

sb-prob3166.problem

A conducting rod of length 1.35m moves with a velocity 45.3m/s parallel to a long wire carrying a steady current of 29.2 mA. The axis of the rod is maintained perpendicular to the wire with the near end a distance 7.33cm away. Calculate the magnitude of the emf induced in the rod.

Tries 0/10

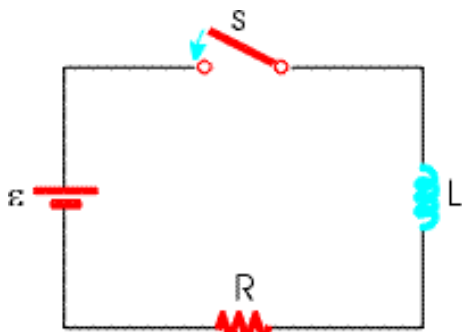
sb-prob3201.problem

A coil has an inductance of 3.00E-3H, and the current through it changes from 2.00E-1A to 1.50E+0A in a time of 2.00E-1s. Find the magnitude of the average induced emf in the coil during this time.

Tries 0/10

sb-prob3219a.problem

Consider the circuit in the figure below, taking $\mathcal{E} = 9.00\text{V}$, $L = 8.12\text{mH}$, and $R = 4.26\Omega$.



What is the inductive time constant of the circuit?

Tries 0/10

Calculate the current in the circuit $228\mu\text{s}$ after the switch is closed.

Tries 0/10

What is the value of the final steady-state current?

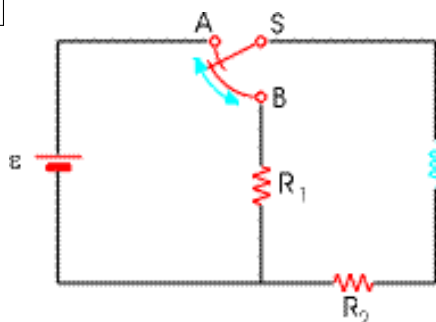
Tries 0/10

How long does it take the current to reach 79.2% its maximum value?

Tries 0/10

sb-prob3228a.problem

One application of an RL circuit is the generation of time-varying high voltage from a low-voltage source. As shown in the figure below, $R_1 = 1320\Omega$, $R_2 = 14.0\Omega$, $L = 2.16\text{H}$ and $\mathcal{E} = 12.0\text{V}$. What is the current in the inductor a long time after the switch has been in position A ?



Tries 0/10

Now the switch is thrown quickly from A to B . Compute the initial voltage across R_1 .

Tries 0/10

Compute the initial voltage across R_2 .

Tries 0/10

What is the initial voltage across the inductor?

Tries 0/10

How much time elapses before the voltage across the inductor drops to $\mathcal{E} = 12\text{V}$?

Tries 0/10

sb-prob3232a.problem

The magnetic field inside a superconducting solenoid is 4.20T . The solenoid has an inner diameter of 6.13cm and a length of 27.7cm . Determine the magnetic energy density in the field.

Tries 0/10

Calculate the energy stored in the magnetic field within the solenoid.

Tries 0/10

sf-prob2028.problem

A 114-turn square wire coil with sides of length $L = 19.7\text{cm}$ rotates about a vertical axis at 1790rpm , as indicated in the figure below. The horizontal component of the Earth's magnetic field at the location of the loop is $2.31\text{E-}5\text{T}$. Calculate the maximum emf induced in the coil by the Earth's field.

Tries 0/10

